

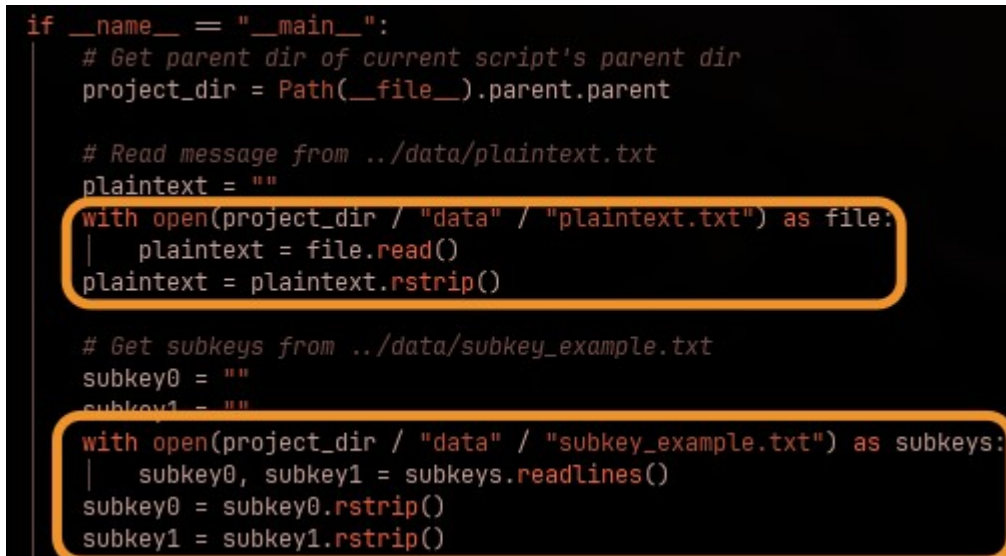
System Information:

- Operating System: Arch Linux ([Omarchy](#))
- Language: Python 3.14.0

Instructions:

1. Download zip folder
2. Unzip
3. Open terminal in the **PARENT** directory of the project – if the project was downloaded to your downloads folder, open the terminal at Downloads/ and run the command in step 4 from there
4. Use “python -m aes_m15219531.build.aes_first_round”

Screenshots:

A screenshot of a terminal window with a dark background and light-colored text. The code is a Python script. Two sections of the code are highlighted with orange rounded rectangles. The first section shows the opening and reading of 'plaintext.txt'. The second section shows the opening and reading of 'subkey_example.txt'.

```
if __name__ == "__main__":  
    # Get parent dir of current script's parent dir  
    project_dir = Path(__file__).parent.parent  
  
    # Read message from ../data/plaintext.txt  
    plaintext = ""  
    with open(project_dir / "data" / "plaintext.txt") as file:  
        | plaintext = file.read()  
        plaintext = plaintext.rstrip()  
  
    # Get subkeys from ../data/subkey_example.txt  
    subkey0 = ""  
    subkey1 = ""  
    with open(project_dir / "data" / "subkey_example.txt") as subkeys:  
        | subkey0, subkey1 = subkeys.readlines()  
        subkey0 = subkey0.rstrip()  
        subkey1 = subkey1.rstrip()
```

Figure 1: Reading *plaintext.txt* and *subkey_example.txt* from the *data/* directory in *build/aes_first_round.py*

```
def aes_first_round(plaintext_input, subkey0_input, subkey1_input):
    # Convert to ASCII (ord), then to hexadecimal (hex), and then to hex literal (int)
    text = [int(hex(ord(character)), 16) for character in plaintext_input]

    # Translate subkeys into usable hex literals
    subkey0 = _string_to_hex(subkey0_input)
    subkey1 = _string_to_hex(subkey1_input)

    # Create 2D array for text, subkey0, and subkey1
    text = _swap_cols_rows(_generate_4x4_matrix(text))
    subkey0 = _swap_cols_rows(_generate_4x4_matrix(subkey0))
    subkey1 = _swap_cols_rows(_generate_4x4_matrix(subkey1))

    # Perform AES calculations
    text = _add_key(text, subkey0)
    text = _sub_bytes(text)
    text = _shift_rows(text)
    text = _mix_columns(text)
    text = _add_key(text, subkey1)

    return text
```

Figure 2: **build/aes_first_round.py** has the function to call the required AddKey (before round 1), SubBytes, ShiftRows, MixColumns, and AddKey (during round 1)

```
def _add_key(text, subkey):    • "_add_key" is not accessed
    """
    text: 4x4 matrix to perform AddKey on
    subkey: subkey to use in the AddKey calculation

    returns: list of len(text)
    """
    flattened_text = [item for sublist in text for item in sublist]
    flattened_subkey = [item for sublist in subkey for item in sublist]

    output_array = []
    count = 0
    while count < len(flattened_text):
        output_array.append(flattened_text[count] ^ flattened_subkey[count])
        count += 1

    output_array = _generate_4x4_matrix(output_array)

    print(f"AddKey: {_hex_convert(output_array)}")
    return output_array
```

Figure 3: **src/aes_functions.py** has the AddKey function, where the 2D arrays are flattened and XOR'ed against each other with the ^ operator

```
def _sub_bytes(text):  ● "_sub_bytes" is not accessed
    """
    text: 4x4 matrix of ints

    returns: list of ints
    """
    sbox = [

    flattened = [item for sublist in text for item in sublist]

    text_strings = []
    for num in flattened:
        text_strings.append(str(hex(num)))

    sbox_key = []
    for num in text_strings:
        # Handle cases like 0x0, 0xe
        if len(num) == 3:
            sbox_key.append([0, int(num[-1], 16)])

        # Handle cases like 0x43, 0xf3
        else:
            sbox_key.append([int(num[-2], 16), int(num[-1], 16)])

    output_array = []
    for key in sbox_key:
        output_array.append(sbox[key[0]][key[1]])

    output_array = _generate_4x4_matrix(output_array)

    print(f"SubBytes: {_hex_convert(output_array)}")
    return output_array
```

Figure 4: *src/aes_functions.py* has the SubBytes function, including the S-Box table (collapsed for readability), table lookup, and 2D matrix reconstruction and output

```
def _shift_rows(text):    • "_shift_rows" is not accessed
    """
    text: 4x4 matrix

    returns: 4x4 matrix
    """
    output_array = []

    index = 0
    while index < len(text):
        output_array += [text[index][index:] + text[index][:index]]
        index += 1

    print(f"ShiftRows: {_hex_convert(output_array)}")
    return output_array
```

Figure 5: *src/aes_functions.py* has the *ShiftRows* function that adjusts row 0 by 0 places, row 1 left once, row 2 left twice, and row 3 left 3 times

```

def _mix_columns(text):
    """
    text: 4x4 matrix

    returns: 4x4 matrix
    """
    mix_cols_constant = [[2, 3, 1, 1], [1, 2, 3, 1], [1, 1, 2, 3], [3, 1, 1, 2]]

    # Init result matrix with 0's
    output_array = [[0 for _ in range(4)] for _ in range(4)]

    for i in range(4):
        for j in range(4):
            val = 0
            for k in range(4):
                val ^= _gf_mul(mix_cols_constant[j][k], text[k][i])
            output_array[j][i] = val & 0xFF

    print(f"MixColumns: {_hex_convert(output_array)}")
    return output_array

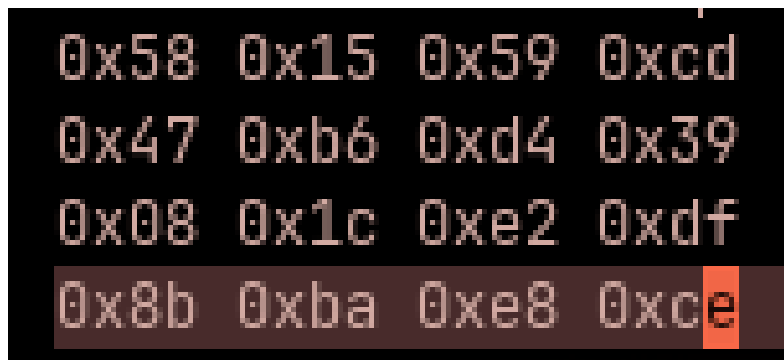

def _gf_mul(num1, num2):
    """
    num1: hex literal
    num2: hex literal

    returns: hex literal
    """
    output = 0

    while num2:
        if num2 & 1:
            output = output ^ num1
        num1 <<= 1
        if num1 & 0x100:
            num1 = (num1 ^ 0x11B) & 0xFF
        num2 >>= 1
    return output & 0xFF

```

Figure 6: `src/aes_functions.py` has the `MixColumns` function (and its helper) that performs the necessary `MixColumns` calculation on the text matrix



```
0x58 0x15 0x59 0xcd
0x47 0xb6 0xd4 0x39
0x08 0x1c 0xe2 0xdf
0x8b 0xba 0xe8 0xce
```

Figure 7: ***data/result.txt*** contains the resulting ciphertext